

Retail Sales Performance and Customer Analytics Dashboard

1. Executive Summary:

This project develops an interactive Power BI dashboard analyzing retail sales performance and customer behavior for an online retail business, using a real transactional dataset of 534,000+ records spanning December 2010 to December 2011. The dashboard combines SQL-based analysis, Python-driven data cleaning and forecasting, and Power BI visualization to translate raw transaction data into actionable insight for sales, marketing, and customer strategy decisions. The end deliverable is a multi-page, cross-filterable dashboard covering sales performance, customer segmentation, and forward-looking sales forecasts.

2. Problem Statement:

Background: Retail businesses generate large volumes of transactional data daily, but this data is often scattered and difficult to translate into timely, actionable insight. Stakeholders need clear visibility into what is selling, who is buying, and where growth or risk is emerging — without manually querying raw transaction logs.

Objective: Develop an interactive dashboard to monitor sales performance, segment customers by value, and forecast near-term revenue, supporting faster and better-informed business decisions.

Scope: Initial focus on overall sales performance (revenue, orders, products, returns), customer analytics (RFM-based segmentation), and short-term sales forecasting, using data spanning all product categories and the retailer's full customer base across multiple countries.

3. Data Sources:

Primary Data: Transaction-level retail sales data (Online Retail II dataset): invoice number, product/stock code, description, quantity, unit price, invoice date, customer ID, and country — approximately 534,000 cleaned transaction records across 4,300+ customers and 3,900+ products.

Secondary Data: None required for the initial phase; a future iteration could incorporate external market or seasonal benchmark data for comparative analysis.

4. Methodology:

Data Integration

- Extract raw transactional data and clean it in Python (pandas): handle missing values, remove duplicates, standardize data types, and flag returns/cancellations.
- Load the cleaned dataset into Power BI via CSV import, and structure supporting tables (calendar/date table, customer segmentation table) for relational analysis.

Analysis

- Use SQL to answer core business questions: monthly revenue trends, top products, revenue by country, average order value, and returns impact.
- Perform RFM (Recency, Frequency, Monetary) customer segmentation in Python to classify customers into value tiers (VIP/Champions, Loyal, Potential Loyalists, At Risk/Churned, Needs Attention).
- Apply time-series forecasting (Holt-Winters exponential smoothing) to project sales for the next reporting period.

Dashboard Design

- Design KPI cards, trend charts, and ranking visuals to surface the answers to each business question clearly.
- Build a dedicated Customer Analytics page using RFM segment data to visualize value distribution across the customer base.

Interactivity

- Implement cross-filtering across all visuals so that selecting any product, country, or segment dynamically updates KPIs and related charts.
- Apply Top-N filtering on ranked visuals (top products, top customers) to keep views focused and readable.
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5. Expected Outcomes:

- An interactive, multi-page dashboard providing real-time insight into sales performance, customer value, and forecasted trends.
- Clear identification of top-performing products, markets, and a data-driven customer segmentation model distinguishing high-value from at-risk customers.
- A reusable analytical pipeline (SQL + Python + Power BI) that can be adapted to future datasets or extended with new business questions.

6. Tools and Technologies:

- Power BI — dashboard development, DAX measures, and interactive visualization.

- Python (pandas, matplotlib, seaborn, statsmodels) — data cleaning, exploratory analysis, RFM segmentation, and forecasting.
- SQL — business-question-driven querying and validation of key metrics.
- Excel — supplementary data checks and quick validation during development.

7. Risks and Challenges:

- Data quality issues in the raw dataset (missing customer IDs, cancelled/returned orders, duplicate records) required careful cleaning to avoid skewed metrics.
- Ensuring consistency between Python-calculated metrics and Power BI DAX measures, particularly around gross vs. net revenue and order counts including or excluding cancellations.
- Data model relationship and type-matching issues (e.g., date/time granularity mismatches between tables) required troubleshooting to ensure accurate cross-filtering.

8. Conclusion:

This project delivers a practical, end-to-end analytics solution — from raw transactional data to an interactive, decision-ready dashboard. By combining SQL, Python, and Power BI, it demonstrates the full workflow of a business/data analyst: translating a business problem into structured requirements, validating findings across tools, and presenting insight in a way stakeholders can act on directly. The resulting dashboard is designed to be clear, accurate, and adaptable to future business questions.